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faster with greater force through the afferent arteriole and leaves the glomerulus through the efferent arterioles, much slower. This force is because of the difference in sizes between the afferent and efferent arteriole (afferent arteriole is wider than efferent arteriole) and glomerular hydrostatic pressure which is around 55mm Hg.

Kidneys produce about 180L of glomerular filtrate in 24 hours. The molecules such as water, glucose, amino acids and nitrogenous substances pass freely from the blood into the glomerulus. Molecules larger than 5nm are barred from entering the tubule. Glomerular pressure is the chief force that pushes water and solutes out of the blood and across the filtration membrane. The glomerular blood pressure (approximately 55 mmHg) is much higher than in other capillary beds. The two opposing forces are contributed by the plasma proteins in the capillaries. These includes, colloidal osmotic pressure (30 mmHg) and the capsular hydrostatic pressure (15 mmHg) due to the fluids in the glomerular capsule. The net filtration pressure of 10 mmHg is responsible for the renal filtration.

Net filtration Pressure 5 Glomerular hydrostatic pressure 2 (Colloidal osmotic pressure 1 Capsular hydrostatic pressure).

Net filtration pressure 5 55 mmHg 2 (30 mmHg 1 15 mmHg) 5 10mmHg

The effective glomerular pressure of 10 mmHg results in **ultrafiltration.** Glomerular filtration rate (GFR) is the volume of filtrate formed min^{21} in all nephrons (glomerulus) of both the kidneys. In adults the GFR is approximately 120-125mL/min. Blood from the glomerulus is passed out through the efferent arteriole. The smooth muscle of the efferent arteriole contract resulting in vasoconstriction. Table 8.1 shows the relative concentrations of substances in the blood plasma and the glomerular filtrate. The glomerular filtrate is similar to blood plasma except that there are no plasma proteins.

Renal clearance is a parameter that reflects the amount of solute passing from the plasma to the urine in a given period of time. If the renal clearance is equal to the GFR it means that there is efficient filtration with little reabsorption and secretion. It is one of the parameters used to identify the efficiency of the kidney.

Table 8.1 Concentration of substances in the blood plasma and in the glomerular filtrate

Substance	Concentration in blood Plasma/g dm^{-3}	Concentration in glomerular filtrate/g dm^{-3}
Water	900	900
Proteins	80.0	0.05
Aminoacids	0.5	0.5
Glucose	1.0	1.0
Urea	0.3	0.3
Uric acid	0.04	0.04
Creatinine	0.01	0.01
Inorganic ions (mainly Na^1, K^1 and Cl^2)	7.2	7.2

A person with cirrhosis of the liver has lower than normal levels of plasma proteins and higher than normal GFR. Explain why a decrease in plasma protein would increase GFR.